

Computation Graphs, Backpropagation and Autodiff

How gradients flow through neural networks

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ES 667 · Deep Learning · IIT Gandhinagar

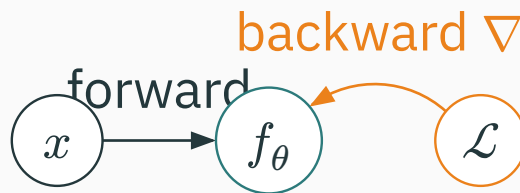
Backprop vocabulary

The goal: two passes over one graph

Training is a **forward** pass and a **backward** pass over the same computation graph:

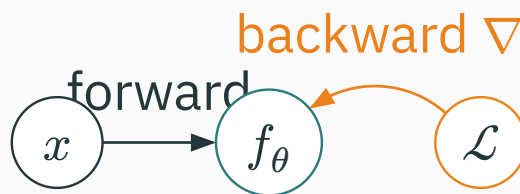
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forward: $x \rightarrow \mathcal{L}$ • **backward:** $\mathcal{L} \rightarrow \nabla_\theta \mathcal{L}$

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Every update step is

$$\theta_{t+1} = \theta_t - \eta \nabla_{\theta} \mathcal{L}(\theta_t).$$

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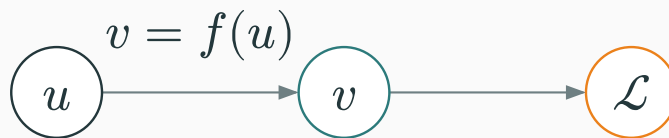
We need **all** P partial derivatives $\partial \mathcal{L} / \partial \theta_j$ — cheaply, and exactly. Backprop delivers them in **one backward pass**.

One local node

Zoom in on a single edge of the graph: $u \rightarrow v \rightarrow \mathcal{L}$.

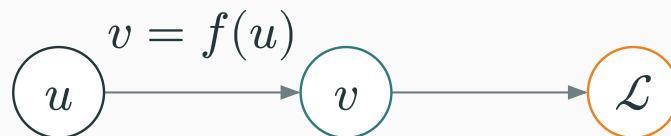
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We want $\bar{u} := \partial \mathcal{L} / \partial u$, and we are **given** $\bar{v} := \partial \mathcal{L} / \partial v$ from the node ahead.

The chain rule splits the work into two pieces we already have:

$$\underbrace{\partial \frac{\mathcal{L}}{\partial} u}_{\text{downstream}} = \underbrace{\partial \frac{\mathcal{L}}{\partial} v}_{\text{upstream}} \cdot \underbrace{\partial \frac{v}{\partial} u}_{\text{local}}$$

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Upstream comes from the node ahead. **Local** is the derivative of **this** node alone. Their product flows **downstream**.

The rule, in bar notation

Write $\bar{u} = \partial \mathcal{L} / \partial u$. Then every node obeys

$$\bar{u} = \bar{v} \cdot \frac{\partial v}{\partial u}.$$

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The node only needs to know its **own** local derivative $\partial v / \partial u$. It multiplies whatever gradient $\bar{v}$ arrives from above — no global bookkeeping.

Example: a square node

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downstream $\bar{u} = 42$

The node did **one** thing: multiply the incoming $\bar{v} = 7$ by its own local slope $2u = 6$.

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+ **copies** the upstream gradient to every input

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If $a = 2, b = 5, \bar{v} = 3$: then $\bar{a} = 3 \cdot 5 = 15$ and $\bar{b} = 3 \cdot 2 = 6$.

Node $v = \varphi(u)$ (elementwise nonlinearity). Local derivative $\partial v / \partial u = \varphi'(u)$.

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sigmoid σ : $\varphi'(u) = \sigma(u)(1 - \sigma(u))$

ReLU: $\varphi'(u) = \mathbb{1}[u > 0]$

The rule table — memorize this

Forward op	Local grad	Backward update
$v = a + b$	$1, 1$	$\bar{a} += \bar{v}, \bar{b} += \bar{v}$
$v = ab$	b, a	$\bar{a} += \bar{v}b, \bar{b} += \bar{v}a$
$v = u^2$	$2u$	$\bar{u} += \bar{v} \cdot 2u$
$v = \varphi(u)$	$\varphi'(u)$	$\bar{u} += \bar{v}\varphi'(u)$

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$v = \varphi(u)$	$\varphi'(u)$	$\bar{u} += \bar{v}\varphi'(u)$

Use +=, not = — a variable feeding several nodes accumulates gradient from **all** of them.

Gradients accumulate at branches

If x influences $\mathcal{L}$ through several intermediate variables $u_1, \dots, u_k$:

$$\frac{\partial \mathcal{L}}{\partial x} = \sum_{i=1}^k \frac{\partial \mathcal{L}}{\partial u_i} \frac{\partial u_i}{\partial x}.$$

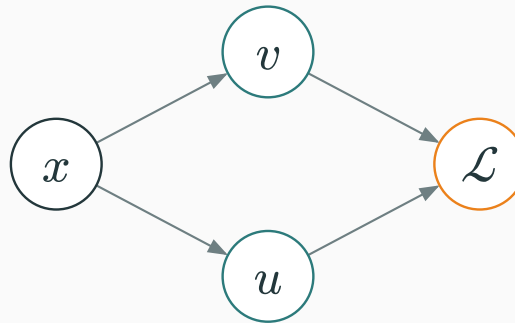
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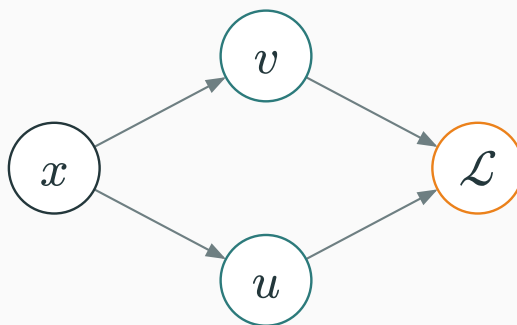
gradients from every path add

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$$\bar{x} = \bar{x}_{\text{via } u} + \bar{x}_{\text{via } v}.$$

Worked branch example

Let $u = x^2$, $v = 3x$, $\mathcal{L} = u + v = x^2 + 3x$.

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Let $u = x^2$, $v = 3x$, $\mathcal{L} = u + v = x^2 + 3x$. By direct calculus:

$$\frac{d\mathcal{L}}{dx} = 2x + 3, \quad \text{at } x = 4: \quad 2(4) + 3 = 11.$$

Worked branch example

Let $u = x^2$, $v = 3x$, $\mathcal{L} = u + v = x^2 + 3x$. By direct calculus:

$$\frac{d\mathcal{L}}{dx} = 2x + 3, \quad \text{at } x = 4: \quad 2(4) + 3 = 11.$$

Now let us get the **same** answer by backprop.

Forward at $x = 4$: $u = 16$, $v = 12$, $\mathcal{L} = 28$.

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$$\overline{x}_{\text{via } u} = \overline{u} \cdot 2x = 1 \cdot 8 = 8, \quad \overline{x}_{\text{via } v} = \overline{v} \cdot 3 = 3.$$

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$$\overline{x}_{\text{via } u} = \overline{u} \cdot 2x = 1 \cdot 8 = 8, \quad \overline{x}_{\text{via } v} = \overline{v} \cdot 3 = 3.$$

$$\overline{x} = 8 + 3 = 11. \quad \checkmark$$

Why += matters

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In PyTorch, `.grad` **accumulates** across `backward()` calls — that is exactly this rule. You must `optimizer.zero_grad()` each step, or gradients from old steps pile up.

Why is a branch node's backward update written with += rather than =?

A. It makes the forward pass faster

B. The variable receives a contribution from every downstream path

C. It changes multiplication into addition

D. Autodiff cannot store scalar values

Correct: B — The variable receives a contribution from every downstream path

Why: The chain rule sums all paths from the variable to the loss; assignment would discard an earlier contribution.

The central worked example

Predict $\hat{y} = wx + b$, squared-error loss $\mathcal{L} = (\hat{y} - y)^2$.

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$$x = 3, \quad w = 2, \quad b = 1, \quad y = 10.$$

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$x = 3$, $w = 2$, $b = 1$, $y = 10$.

Four scalars, one loss. We will get $\partial\mathcal{L}$ w.r.t. **all four** by hand, then check with PyTorch.

Break into primitives

Decompose the loss into elementary nodes:

$$m = wx, \quad a = m + b, \quad e = a - y, \quad \mathcal{L} = e^2.$$

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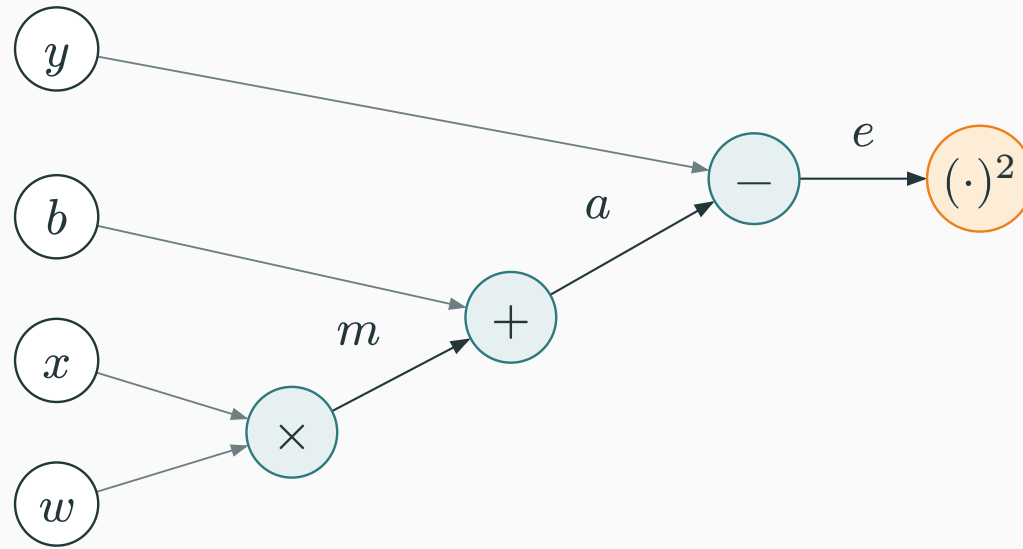
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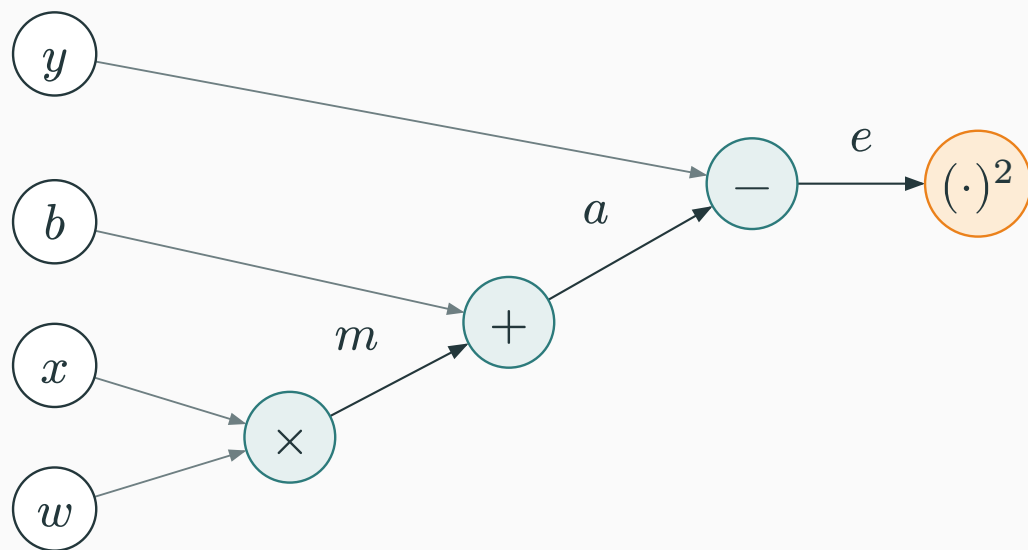
$\times$, then $+$, then $-$, then $(\cdot)^2$

Each is one of the nodes from our rule table.

The computation graph



The computation graph



inputs (ink) $\rightarrow$ ops $\times, +, -$ (teal) $\rightarrow$ loss $(\cdot)^2$ (orange)

Node	Rule	Value
$m = wx$	$2 \cdot 3$	6
$a = m + b$	$6 + 1$	7
$e = a - y$	$7 - 10$	-3
$\mathcal{L} = e^2$	$(-3)^2$	9

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$\mathcal{L} = 9$ — now run the graph **backward**

Seed the output: $\overline{\mathcal{L}} = \partial \mathcal{L} / \partial \mathcal{L} = 1.$

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$$\bar{e} = \overline{\mathcal{L}} \cdot 2e = 1 \cdot 2(-3) = -6.$$

Node $e = a - y$, locals $\partial e / \partial a = 1$, $\partial e / \partial y = -1$:

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$$\bar{a} = \bar{e} \cdot 1 = -6$$

$$\bar{y} = \bar{e} \cdot (-1) = 6$$

The target y already has a gradient — we would use $\bar{y}$ if y were itself a learned quantity.

Node $a = m + b$ **copies** its gradient:

$$\overline{m} = \overline{a} = -6, \quad \overline{b} = \overline{a} = -6.$$

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$$\bar{m} = \bar{a} = -6, \quad \bar{b} = \bar{a} = -6.$$

So $\bar{b} = -6$ is one of our final answers.

Backward: the multiplication node

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$$\bar{w} = -18, \quad \bar{x} = -12$$

$\partial \mathcal{L} / \partial w$	$\partial \mathcal{L} / \partial b$	$\partial \mathcal{L} / \partial x$	$\partial \mathcal{L} / \partial y$
-18	-6	-12	6

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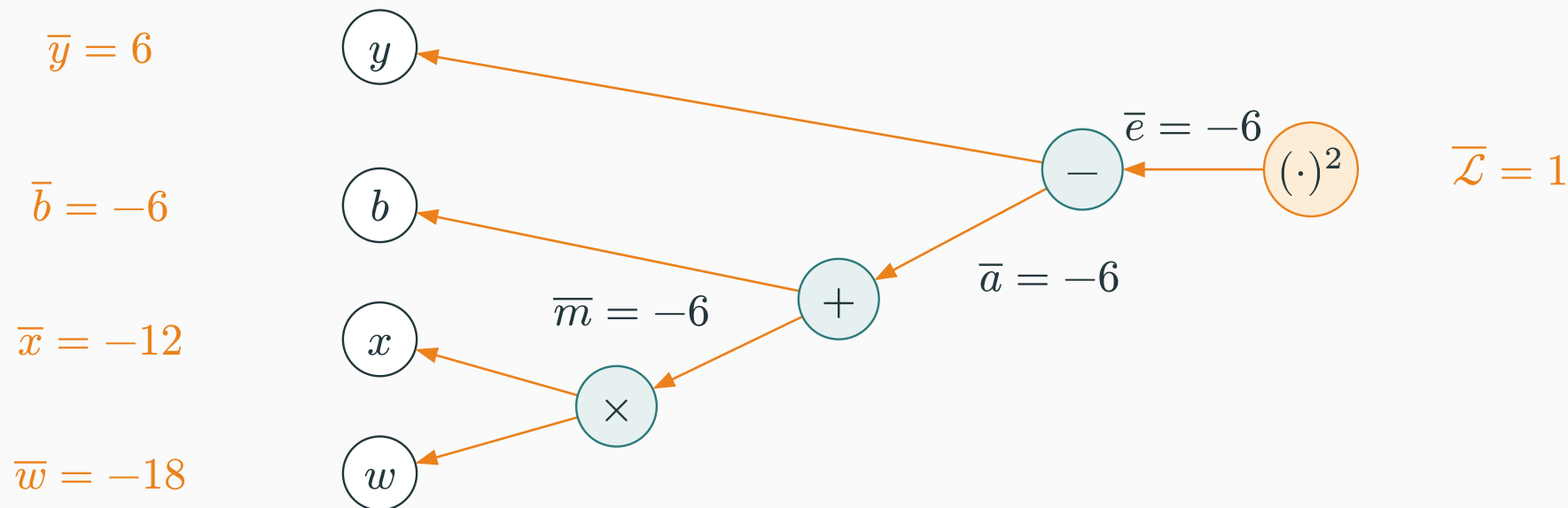
One backward sweep produced **all four** partial derivatives — no formula re-derivation per parameter.

The backward pass, visualized

Seed $\overline{\mathcal{L}} = 1$ at the loss; each **orange arrow** carries **upstream** $\times$ **local** one node to the left:

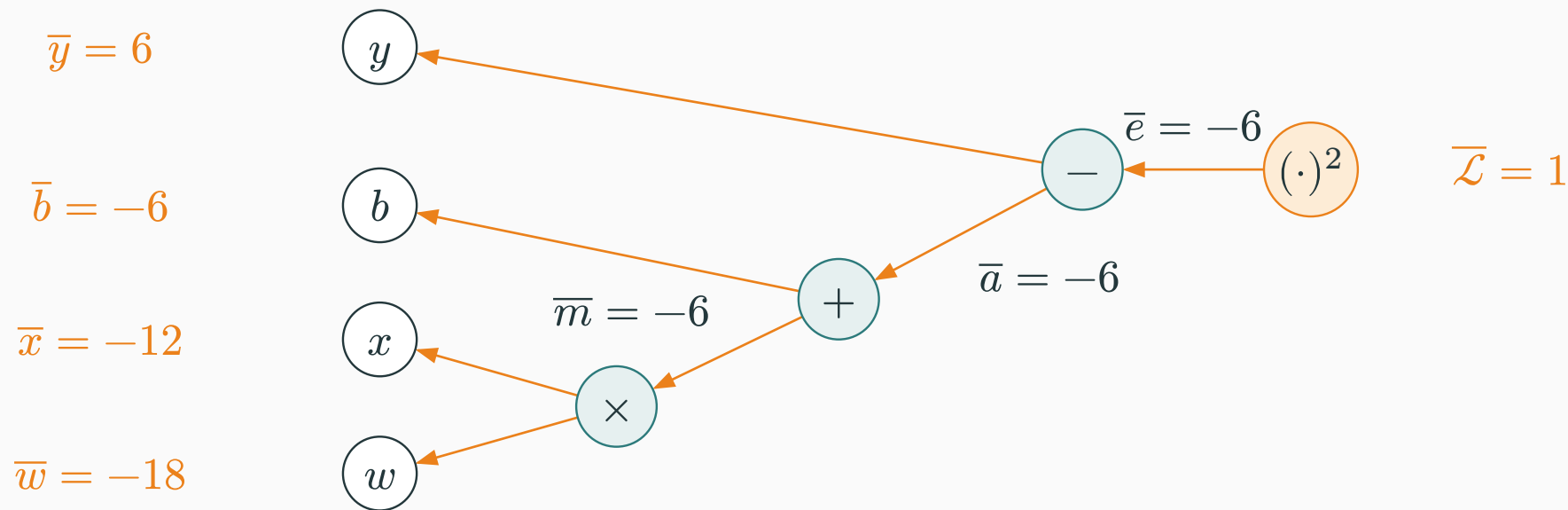
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same graph as the forward pass — every arrow reversed, adjoints filling in right to left

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backprop = calculus — but mechanical & reusable

One gradient-descent step

Take $\eta = 0.1$ and update:

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Target is $y = 10$; we jumped from 6 to 13 — we **overshot**. Lecture 4 is about choosing η and the update rule so this is stable.

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/autograd

Build a scalar computation graph, run the forward pass, then click **backward** to watch each adjoint $\bar{u}$ fill in — exactly the $(wx + b - y)^2$ example above.

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Build a scalar computation graph, run the forward pass, then click **backward** to watch each adjoint $\bar{u}$ fill in — exactly the $(wx + b - y)^2$ example above.

Trace which local rule changes when the final squared-error operation is replaced with $|e|$.

From scalar graph to a neuron

A single neuron

Vector input, one output:

$$z = w^\top x + b, \quad a = \varphi(z), \quad \mathcal{L} = \ell(a, y).$$

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same three nodes: affine → activation → loss

Let $\delta := \bar{z} = \partial \mathcal{L} / \partial z$. Through $a = \varphi(z)$:

$$\delta = \bar{a} \cdot \varphi'(z), \quad \text{where } \bar{a} = \frac{\partial \mathcal{L}}{\partial a}.$$

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δ is the **error signal at the pre-activation**. Everything below the neuron is expressed through this one scalar.

Backprop through the affine part

D DERIVATION

Through $z = w^\top x + b$ with $\bar{z} = \delta$:

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$$\nabla_w \mathcal{L} = \delta x, \quad \frac{\partial \mathcal{L}}{\partial b} = \delta, \quad \bar{x} = \delta w.$$

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Parameter gradient = error δ times the **input** x . This “ δ times input” shape reappears at every layer.

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$$\bar{z} = a - y \text{ — prediction minus target}$$

For multi-class, $p = \text{softmax}(z)$ and $\mathcal{L} = -\log p_y$:

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Choosing the loss to **match** the output nonlinearity makes the output-layer gradient trivially simple.

The dense layer

A dense layer maps $x \in \mathbb{R}^n$ to $z \in \mathbb{R}^m$:

$$z = Wx + b, \quad W \in \mathbb{R}^{m \times n}, \quad b \in \mathbb{R}^m.$$

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Read the shapes off the equation: W has one row per output, one column per input.

Backward through a dense layer

Given upstream $\bar{z} \in \mathbb{R}^m$:

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$$\nabla_W \mathcal{L} = \bar{z} x^\top \quad (m \times n), \quad \nabla_b \mathcal{L} = \bar{z}, \quad \bar{x} = W^\top \bar{z} \quad (n).$$

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$$\nabla_W = \bar{z} x^\top \cdot \bar{x} = W^\top \bar{z}$$

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Summing rows of $\bar{Z}$ for ∇_b is just the branch rule: b is **shared** across all N examples, so its gradients accumulate.

Quantity	Shape	Must match
W	$m \times n$	∇_W
b	m	∇_b
x	n	$\bar{x}$
z	m	$\bar{z}$

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gradient shape = parameter shape — always

Backprop through a two-layer MLP

$$z^{(1)} = W_1 x + b_1, \quad h = \varphi(z^{(1)}),$$

$$z^{(2)} = W_2 h + b_2, \quad p = \text{softmax}(z^{(2)}), \quad \mathcal{L} = -\log p_y.$$

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affine → activation → affine → softmax → CE

The forward graph



The forward graph



backprop reverses every arrow, right to left

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Then, using the dense-layer rules:

$$\nabla_{W_2} \mathcal{L} = \bar{z}^{(2)} h^\top, \quad \bar{h} = W_2^\top \bar{z}^{(2)}.$$

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$\odot$ is elementwise (Hadamard) — the activation Jacobian is diagonal.

Forward	Backward
$z^{(1)} = W_1 x + b_1$	$\bar{z}^{(1)} = \bar{h} \odot \varphi'(z^{(1)})$
$h = \varphi(z^{(1)})$	$\nabla_{W_1} = \bar{z}^{(1)} x^\top, \nabla_{b_1} = \bar{z}^{(1)}$
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same two moves at every layer: $\delta x^\top$ for weights, $W^\top \delta$ downstream

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/autograd

Step through a small MLP's forward pass, then watch $\bar{z}^{(2)} = p - y$ propagate back through W_2 , the activation, and W_1 — the adjoints light up layer by layer.

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Trace the same upstream adjoint into the parameter and input branches of the affine layer.

For $z = Wx + b$, which pair of gradients uses the same upstream adjoint $\bar{z}$?

A. $\nabla_W = \bar{z}x^\top$ and $\bar{x} = W^\top \bar{z}$

B. $\nabla_W = W^\top x$ and $\bar{x} = \bar{z}x^\top$

C. $\nabla_W = x$ and $\bar{x} = W$

D. $\nabla_W = 0$ and $\bar{x} = 0$

Correct: A — $\nabla_W = \bar{z}x^\top$ and $\bar{x} = W^\top \bar{z}$

Why: Both branches apply the chain rule to the same output adjoint; they differ only in the local derivative of the affine operation.

Automatic differentiation

Backprop is reverse-mode autodiff

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it is reverse-mode automatic differentiation: the chain rule applied to the executed program

Why reverse mode?

A loss is a map $f : \mathbb{R}^P \rightarrow \mathbb{R}$: **many** inputs, **one** output.

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$P \approx 10^9$ **inputs, one scalar loss $\Rightarrow$ reverse mode wins**

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you write only the forward pass — the backward pass is generated

```
import torch
x = torch.tensor(3.0)
w = torch.tensor(2.0, requires_grad=True)
b = torch.tensor(1.0, requires_grad=True)
y = torch.tensor(10.0)

L = (w*x + b - y)**2      # forward:  L = 9
L.backward()              # reverse-mode autodiff

print(L.item())           # 9.0
print(w.grad, b.grad)     # tensor(-18.)  tensor(-6.)
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matches our hand computation exactly

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#import "@local/autodiff:0.1.0" as ad
#let L = ad.expr("(w*x + b - y)^2", ("w", "x", "b", "y"))
#ad.value(L, (2, 3, 1, 10)) // the loss
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computed live in this slide: $\mathcal{L} = 9$, $\bar{w} = -18$, $\bar{x} = -12$, $\bar{b} = -6$, $\bar{y} = 6$ — the same four numbers, by hand, by PyTorch, and by our own graph.

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Forget `zero_grad()` and you sum gradients across steps — the same += branch rule, now biting you.

Gradient checking & gradient flow

To **debug** an analytic gradient, compare against a central difference:

$$\frac{\partial \mathcal{L}}{\partial \theta_j} \approx \frac{\mathcal{L}(\theta + \varepsilon e_j) - \mathcal{L}(\theta - \varepsilon e_j)}{2\varepsilon}.$$

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Great for **unit tests** ($\varepsilon \approx 10^{-5}$), far too slow for training — one loss evaluation **per parameter**.

Stack L layers $h_0 \rightarrow h_1 \rightarrow \dots \rightarrow h_L$. The chain rule multiplies Jacobians:

$$\frac{\partial \mathcal{L}}{\partial h_0} = \frac{\partial \mathcal{L}}{\partial h_L} \prod_{\ell=1}^L \frac{\partial h_\ell}{\partial h_{\ell-1}}.$$

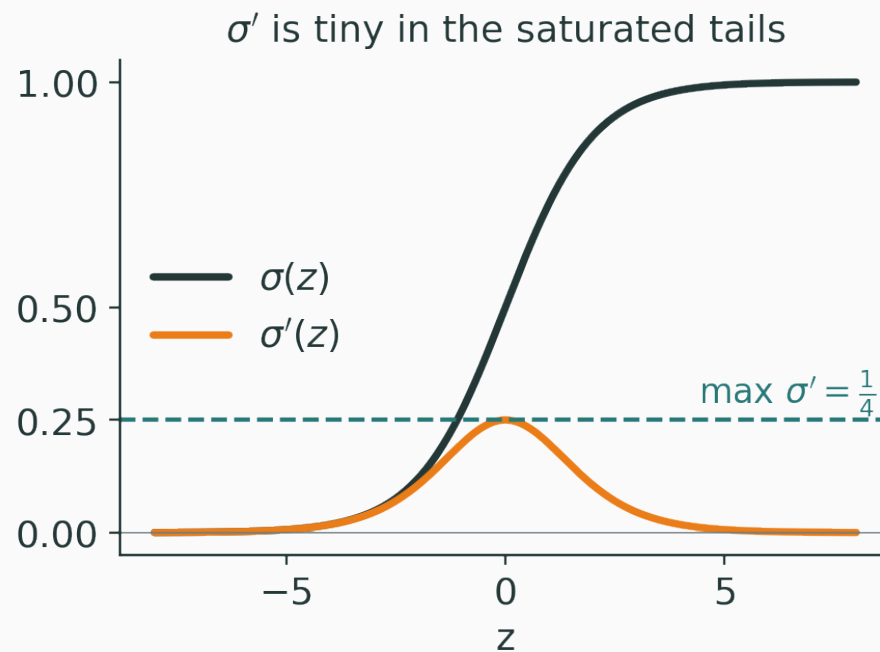
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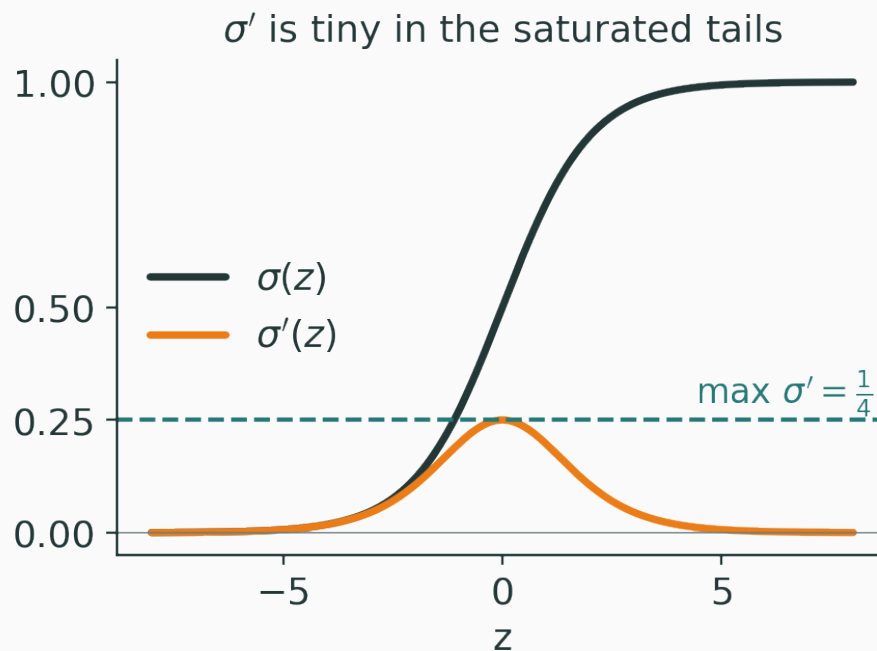
a product of L terms — it can vanish or explode

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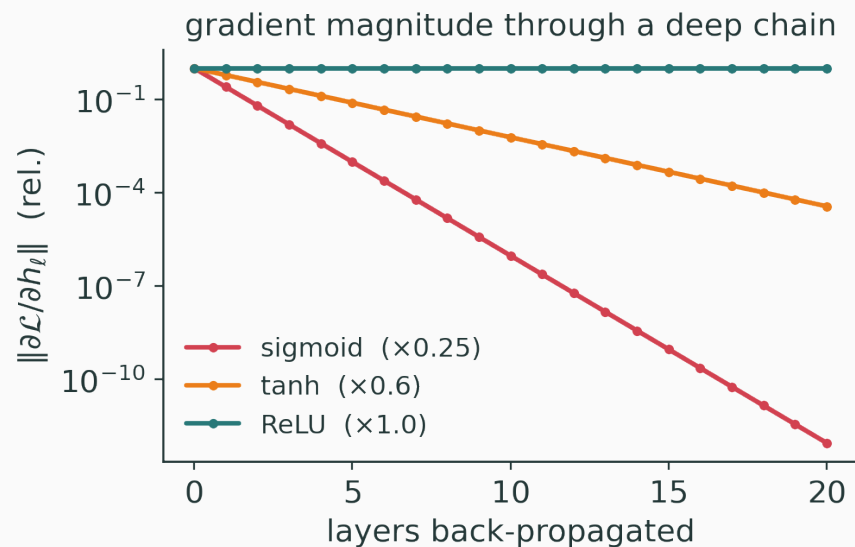
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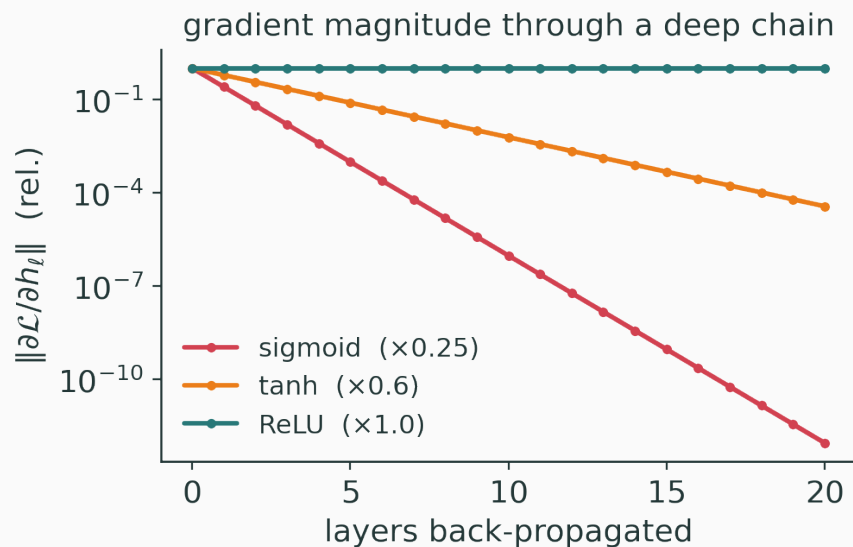
each sigmoid layer multiplies the gradient by at most $1/4$

Multiply many small local derivatives $\rightarrow$ the signal **decays** with depth:

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Sigmoid decays fast; ReLU (local slope 0 or 1) keeps gradients alive.

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/vanishing-gradients

Sweep depth and activation function; watch the gradient magnitude at layer 0 collapse for sigmoid and survive for ReLU.

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Sweep depth to see repeated small local slopes rapidly erase an early-layer gradient.

Summary

Backprop in one sentence

Seed $\bar{\mathcal{L}} = 1$, then walk the graph **backward**, at each node computing

$$\text{downstream} = \text{upstream} \times \text{local},$$

and **adding** gradients wherever a variable branched.

Setting	Backward rule
scalar node	$\bar{u} = \bar{v} \cdot \partial v / \partial u$
dense layer	$\nabla_W = \bar{z} x^\top, \quad \bar{x} = W^\top \bar{z}$
activation	$\bar{z} = \bar{a} \odot \varphi'(z)$
softmax + CE	$\bar{z} = p - y$
branch point	gradients add

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Backprop gives the **direction**; optimization decides the **step**.

We can compute $\nabla_{\theta} \mathcal{L}$.

Next: how to **use** it — SGD, momentum, Adam.